MIL-F-279C
6 December 1982
SUPERSEDING
MIL-F-279B
6 May 1969

MILITARY SPECIFICATION

FERROUS SULFIDE, TECHNICAL

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two grades of ferrous sulfide.

1.2 Classification. Ferrous sulfide shall be of the following grades as specified (see 6.2).

Grade A - 75.0 percent FeS in granular form
Grade B - 85.0 percent FeS in stick form

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
RR-S-366	-	Sieve Test
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-96 - Cans, Metal, 28 Gage and Lighter
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-60 - Tape: Packaging, Waterproof

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical characteristics. Ferrous sulfide shall conform to the applicable chemical characteristics of table I when tested as specified therein.

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TABLE I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Grade A	Grade B	
Assay (as FeS), minimum	75.0	85.0	4.2.4.1
Free sulfur, maximum	0.5	--	4.2.4.2

3.2 Granulation (grade A). No less than 99.9 percent by weight of grade A ferrous sulfide shall pass through a No. 100 sieve when tested as specified in 4.2.4.3.

3.3 Appearance (grade B). Grade B ferrous sulfide shall be dark brown cylindrical sticks when tested as specified in 4.2.4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the ferrous sulfide of one grade produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For ferrous sulfide test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 500 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

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TABLE II. Sampling for ferrous sulfide test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	13
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Tape seal incomplete or missing when required
(g) Fiberboard lining and pads missing or not as specified when required
(h) Partitioning not as specified

4.2.3.2 For ferrous sulfide test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.4 Ferrous sulfide tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as FeS).

(a) Ammoniacal cadmium chloride solution. Dissolve 22 g of cadmium chloride, dihydrate in 200 milliliters (ml) of water. Add 480 ml of concentrated ammonium hydroxide solution and dilute to 1 liter with water.

(b) Indicator solution. Mix 1 g of soluble starch with 10 milligrams (mg) of red mercuric iodide and add enough cold water to make a thin paste. Add 200 ml of boiling water and boil for 1 minute while stirring. Cool before use.

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(c) Procedure. Weigh to the nearest 0.1 mg approximately 0.4 g of the specimen into a 200-ml distilling flask fitted with a dropping funnel. Connect the outlet of the distilling flask to two absorption receivers connected in series. Purge the system with a current of nitrogen introduced through a funnel tube. Fill each absorption receiver two-thirds full of ammoniacal cadmium chloride solution prepared as specified in (a). Introduce into the flask, through the dropping funnel, approximately 20 ml of concentrated hydrochloric acid diluted with 30 ml of water. Heat the contents of the flask to boiling. Pass nitrogen through the apparatus until all of the hydrogen sulfide has been driven over, then disconnect the apparatus. Filter the yellow precipitate of cadmium sulfide and wash it with water until the washings are no longer alkaline. Place the filter and precipitate in a 250-ml glass stoppered Erlenmeyer flask and add 105 ml of 0.1N iodine solution. Add 10 ml of concentrated hydrochloric acid, stopper the flask, and shake occasionally until the precipitate is completely dissolved. Pour the solution into each of the absorption tubes, in turn, to dissolve any traces of yellow precipitate adhering to the tubes. Return the solution to the flask, wash out the tubes into the flask with 50 ml of water, and titrate the excess of iodine with 0.1N sodium thiosulfate solution adding indicator solution prepared as specified in (b) near the end of the titration. Calculate the percent by weight ferrous sulfide as follows:

$$\text{Percent by weight ferrous sulfide} = \frac{4.396\ (AB - CD)}{W}$$

where: A = Milliliters of iodine solution added,
B = Normality of iodine solution,
C = Milliliters of sodium thiosulfate solution used,
D = Normality of sodium thiosulfate solution, and
W = Weight of specimen in grams.

4.2.4.2 Free sulfur. Place an extraction thimble in a weighing bottle and heat for at least 1 hour at 100° ± 5°C, stopper, cool to room temperature in a desiccator, and weigh to the nearest milligram. Grind a portion of the specimen so that it passes a No. 200 sieve conforming to RR-S-366. Weigh to the nearest milligram approximately 10 g of the ground specimen into the extraction thimble. Place in a Soxhlet extractor and extract with carbon disulfide for approximately 30 minutes at such a rate that one filling of the chamber takes about 5 minutes. Remove the thimble from the extraction apparatus and place in a beaker over a steam bath in a well-ventilated hood for about 30 minutes in order to evaporate the solvent. Place the thimble in the previously used weighing bottle, heat for 1 hour at 100° to 110°C, stopper the bottle, cool to room temperature in a desiccator, and weigh to the nearest milligram. (The stoppered bottle should be covered with a cloth for protection in case the bottle implodes due to reduced pressure.) Calculate the percent by weight free sulfur as follows:

$$\text{Percent by weight free sulfur} = \frac{100\ (A - B)}{(A - C)}$$

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where: A = Weight of original specimen and thimble in grams,
B = Weight of specimen and thimble after extraction in grams, and
C = Weight of the thimble in grams.

4.2.4.3 Granulation (grade A only). Place a dry No. 100 sieve conforming to RR-S-366 on a weighed bottom pan. Weigh to the nearest 0.01 g approximately 100 g of specimen, transfer to the sieve, and cover the assembly. Shake for 5 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the specimen in the bottom pan and calculate the percent passing.

4.2.4.4 Appearance (grade B only). Visually examine the specimen for form and color.

4.2.5 Container leakage tests.

4.2.5.1 Ferrous sulfide leakage. Place the container in each of the following positions, and leave it in each for a period of 15 minutes.

(a) Upright
(b) Upside down
(c) On one side (or one quadrant)
(d) On one end (or second quadrant)
(e) On other side (or fourth quadrant)

Examine the container after each period for any evidence of leakage.

4.2.5.2 Air leakage. Submerge the container in water heated to a temperature at least 28°C above the identified temperature of the container. While holding the container submerged with the uppermost surface covered by not more than 1 inch of water, observe for at least 15 seconds to detect leakage. The container shall be rotated and observed repeatedly until all of the container has been examined. Total time in hot water should not exceed 8 minutes. Record the locations of any leaks or state "no leaks".

5. PACKAGING

5.1 Unit packing. Ferrous sulfide shall be unit packed level A or industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Grade A ferrous sulfide. A quantity of 1 (+0.05 or -0) lb of grade A ferrous sulfide shall be unit packed in a clean, dry, glass bottle of a minimum size to contain the ferrous sulfide. The bottle shall conform to group A, class 1, type d, style 2 grade 2 of PPP-C-186. The filled bottle shall be closed using closure A and outer seal A, as specified in PPP-C-186. There shall be no evidence of leakage of air or ferrous sulfide when tested as specified in 4.2.5.

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5.1.1.2 Grade B ferrous sulfide. A quantity of 5 lb (+1 or -0 ounces) of grade B ferrous sulfide shall be unit packed in a clean, dry, tin-plate, slip-cover can conforming to type VI of PPP-C-96, of a minimum size to contain the ferrous sulfide. The seam between the cover and the can body shall be completely sealed by tape conforming to class optional of PPP-T-60. There shall be no evidence of leakage of air or ferrous sulfide when tested as specified in 4.2.5.

5.1.2 Industrial. A specified quantity and grade of ferrous sulfide shall be unit packed in accordance with MIL-STD-1188 (see 6.2).

5.2 Packing. Ferrous sulfide shall be packed level A, B or industrially as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Grade A ferrous sulfide. Twenty-four bottles of grade A ferrous sulfide, unit packed as specified in 5.1, shall be packed in two layers, each having three rows of four bottles, in a close-fitting wood box. The wood box shall conform to PPP-B-585, class 3, style optional for a type I load, or to PPP-B-621 class 2, style 4 for a type 2 load. All inside faces of the box shall be lined with fiberboard, and each bottle shall be inserted in a close-fitting cell formed by full bottle height fiberboard half-slotted partitions. A fiberboard pad shall be placed between the two layers of bottles. The fiberboard shall conform to PPP-F-320, grade V3c or W5c. Each box shall be closed as specified in the box specification, and the box conforming to PPP-B-621 shall be reinforced by strapping as specified therein.

5.2.1.2 Grade B ferrous sulfide. Four 5-lb cans of grade B ferrous sulfide shall be packed in a horizontal two-can by two-can pattern in a wood box, lined and partitioned as specified above for the grade A material.

5.2.2 Level B. Unit packs of grade A and grade B ferrous sulfide shall be packed level B in the same manner as for level A except that the box shall be fiberboard conforming to PPP-B-636, grade V3c.

5.2.3 Industrial. Ferrous sulfide shall be packed in accordance with MIL-STD-1188.

5.3 Unitization. Uniform quantities of level A or B packs of ferrous sulfide shall be palletized in accordance with MIL-STD-147 using load type I and the softwood pallet conforming to NN-P-71, type IV. Industrial packs shall be unitized in a manner to assure carrier acceptance, safe delivery to destination, and stable stacking in storage.

5.4 Marking. Level A and B containers and pallet loads shall be marked in accordance with MIL-STD-129. Industrial containers shall be marked in accordance with MIL-STD-1188. Each container shall also be marked to show the date of manufacture and lot or batch number of the ferrous sulfide.

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6. NOTES

6.1 Intended use. Grade A ferrous sulfide is intended for use in the manufacture of pyrotechnics and for laboratory use. Grade B ferrous sulfide is intended for laboratory use.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of unit packing and packing required (see 5.1 and 5.2)
(c) Grade of ferrous sulfide required (see 1.2)
(d) Quantity of ferrous sulfide to be packed industrially (see 5.1.2)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding off method of ASTM E29.

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5. PROBLEM AREAS

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